Design and Implementation of a General-Purpose Classical AI Logical Reasoning System

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Abstract

This report presents an end-to-end symbolic reasoning system that decides whether a query statement is logically entailed by a knowledge base of facts and rules. Parsing is separated from inference, and every decision is made by a classical reasoning engine; no generative model produces a prediction. The system supports three representations: first-order Horn clauses, deterministic structural causal models (SCMs) and full first-order logic (FOL) syntax trees. It implements forward chaining, backward chaining with unification, A* proof search, Pearl's abduction–action–prediction procedure, and Herbrand-grounded DPLL refutation. On the 3,829 public queries from RuleTaker, CounterBench and FOLIO it obtains 98.5% on RuleTaker, 96.0% on CounterBench and 78.0% on FOLIO (entailed vs. not entailed), with a total execution time of about 1.3 s.

1 Chosen Internal Representation

Input assumption. Following the assignment instructions, the translation from natural language into logic is not the focus of this work. Rule-chaining and causal inputs are parsed with deterministic templates, and for FOLIO the system consumes the supplied first-order annotations (`premises-FOL`, `conclusion-FOL`) as its input logic.

1.1 First-order Horn clauses and atomic facts

For RuleTaker and syllogistic knowledge bases, the knowledge is a set of typed terms (constants c and variables $?x$), atoms $P(t_1, \ldots, t_k)$ and Horn rules

$$P_1(x) \wedge P_2(x) \wedge \cdots \wedge P_m(x) \rightarrow Q(x). \tag{1}$$

Templates such as "Based on A, it is evident that B", "When A, we can conclude B" and "If someone is P and Q, then they are R" are parsed by regular expressions into ground facts and quantified Horn clauses. Queries are answered under the closed-world assumption: an atom that cannot be derived is `False`.

1.2 Structural causal equations (CounterBench)

Deterministic counterfactual questions are represented as assignments in a directed causal graph $G = (V, E)$:

$$v_i := f_i(\mathrm{Pa}(v_i)), \qquad f_i \in \{\wedge, \vee, \neg\}. \tag{2}$$

Statements such as "A and B together cause not C", "V or not T causes S and F" and "A causes B, C and D" are decomposed into one equation per target; a multi-target list inherits the antecedent of the preceding clause. Observations ("We observed X") are stored in an observation map O. Parentless variables that are not observed default to `False`.

1.3 First-order logic syntax trees (FOLIO)

Formulas using $\forall$, $\exists$, $\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$ and $\oplus$ are parsed into a recursive abstract syntax tree. A string-similarity pass (`SequenceMatcher`) aligns predicate and constant names in the conclusion with those of the premises, to absorb small spelling differences between the two annotations.

2 Classical Reasoning Algorithms

All entailment decisions are made by symbolic inference. Forward chaining (RuleTaker), the causal engine (CounterBench) and DPLL refutation (FOLIO) form the evaluated path; backward chaining is used for the natural-language fallback on general knowledge bases, and A* proof search is implemented as an alternative prover but is not used in the benchmark runs.

1. **Forward chaining with Robinson unification.** Starting from the facts F_0, rule antecedents are unified with known ground facts until the query is derived or a fixed point $F_{t+1} = F_t$ is reached.
2. **Backward chaining (DFS with cycle detection).** For a goal q, the engine searches rules whose consequent unifies with q and recursively proves the antecedents, keeping a visited-goal set so cyclic rule graphs terminate.
3. **A* proof search.** A state s is the set of unproven subgoals, $g(s)$ is the number of inference steps and $h(s) = |s \setminus F_0|$ estimates the remaining work.
4. **Abduction–action–prediction.** (i) *Abduction* fixes the observed exogenous variables to their observed values (observations of derived variables are not used); (ii) *action* applies the intervention $\mathrm{do}(X = x)$ by overriding the equations of the intervened variables (several interventions are applied together for joint and nested questions); (iii) *prediction* propagates truth values to evaluate the query.
5. **Herbrand grounding and DPLL refutation.** Quantified formulas are instantiated over a bounded Herbrand universe. KB $\wedge \neg$Query is converted to CNF by a memoized Tseitin transformation and solved by a step-bounded DPLL solver with unit propagation. If it is unsatisfiable (and the KB is satisfiable), the query is entailed. The domain is capped at three constants (those of the conclusion first) and the search stops after 100 nodes, treating the formula as satisfiable, i.e. the query as not entailed.

3 Experiments on the Public Datasets

Label convention. RuleTaker and CounterBench are binary. FOLIO has three labels (`True`, `False`, `Uncertain`); because the task asks for entailment, the system outputs `True` only if the conclusion is entailed and `False` otherwise, so gold labels `False` and `Uncertain` are both treated as "not entailed". Under strict three-way matching the FOLIO accuracy would be lower (46.2% on train, 50.2% on validation), since the system never outputs `Uncertain`.

3.1 Discussion

- **RuleTaker (97.6–99.5%).** Forward chaining answers all 1,625 queries in in about 0.15 s. All 25 errors are false positives: the query is derivable from the rules, but the dataset label is `False`. A likely cause is label noise from the dataset's generator, which may not have registered every derivable rule, although we did not verify this.
- **CounterBench (96.0%).** The structural engine answers 1,000 questions in 0.06 s (about 0.06 ms per question). Accuracy by type is: basic 250/250, conditional 230/250, joint

Table 1: Accuracy and execution time on the public benchmarks (times from one run on the evaluation machine; they vary slightly between runs).

Split	Queries	Correct	Accuracy	Time (s)
`ruletaker_train`	1,000	983	98.30%	0.0916
`ruletaker_test`	250	244	97.60%	0.0229
`ruletaker_ood_test`	375	373	99.47%	0.0376
`counterbench_alpha`	1,000	960	96.00%	0.0609
`FOLIO_train`	1,001	774	77.32%	0.8794
`FOLIO_validation`	203	165	81.28%	0.2014
Overall	3,829	3,499	91.38%	1.2941

240/250, nested 240/250. The remaining errors occur in graphs with observed or intervened variables on negated or "or" branches; we did not isolate a single cause.

- **FOLIO (77.3% train, 81.3% validation).** Always answering `False` would score 61.2% (train) and 64.5% (validation), so the solver is clearly better than the majority baseline. Its errors are asymmetric (Table 2): it rarely asserts an entailment that does not hold, but misses many true entailments, consistent with the three-constant domain cap and the 100-node DPLL budget, both of which push undecided cases towards `False`.

Table 2: FOLIO breakdown (entailed vs. not entailed).

Split	Recall (gold `True`)	Recall (not `True`)	Majority baseline
Train	190/388 (49.0%)	584/613 (95.3%)	61.2%
Validation	41/72 (56.9%)	124/131 (94.7%)	64.5%

4 Examples of Reasoning Traces

Trace 1: unification and forward chaining

KB: Bob is young. Bob is kind. If someone is young, then they are smart. If someone is smart and kind, then they are successful. **Query:** Bob is successful.

- $F_0 = \{\text{young}(\text{bob}), \text{kind}(\text{bob})\}$.
- Step 1: unify young($?x$) with young(bob), $\theta = \{?x \mapsto \text{bob}\}$; derive smart(bob).
- Step 2: unify $\text{smart}(?x) \wedge \text{kind}(?x)$ with $\{\text{smart}(\text{bob}), \text{kind}(\text{bob})\}$; derive successful(bob).
- Prediction: `True`.

Trace 2: RuleTaker OOD, depth 4

Facts: Henry reads books. Sarah goes to the hospital. Henry carries bags. **Rules:** R1: henry carries bags $\rightarrow$ the teacher carries bags; R2: henry carries bags $\rightarrow$ the teacher reads books; R3: the teacher reads books $\rightarrow$ sarah carries bags. **Query:** Sarah carries bags.

- Step 1: *henry carries bags* $\in F_0$ triggers R2 and adds *the teacher reads books*.
- Step 2: that fact triggers R3 and adds *sarah carries bags*.
- Step 3: the goal is in F_2. Prediction: `True`.

Trace 3: CounterBench counterfactual (question 1)

Given: Ziklo causes Blaf, Blaf and Ziklo together cause Trune, Trune causes not Vork, Vork causes Lumbo. **Query:** Would Lumbo occur if not Ziklo instead of Ziklo?

- Equations: $\mathit{blaf} := \mathit{ziklo}$; $\mathit{trune} := \mathit{blaf} \wedge \mathit{ziklo}$; $\mathit{vork} := \neg \mathit{trune}$; $\mathit{lumbo} := \mathit{vork}$.
- Intervention: $\mathrm{do}(\mathit{ziklo} = \mathrm{False})$.
- Propagation: $\mathit{blaf} = \mathrm{False} \Rightarrow \mathit{trune} = \mathrm{False} \Rightarrow \mathit{vork} = \mathrm{True} \Rightarrow \mathit{lumbo} = \mathrm{True}$.
- Prediction: `True` (yes).

Trace 4: FOLIO refutation (example 1175)

Premises: $\forall x\,(\mathrm{Have}(x, \mathrm{typecport}) \rightarrow \neg \mathrm{ProducedBefore}(x, \mathrm{yr2010}))$; $\forall x\,(\mathrm{Monitor}(x) \wedge \mathrm{In}(x, \mathrm{library}) \rightarrow \mathrm{ProducedBefore}(x, \mathrm{yr2010}))$; $\mathrm{Monitor}(\text{l-2021}) \wedge (\mathrm{In}(\text{l-2021}, \mathrm{library}) \oplus \mathrm{Have}(\text{l-2021}, \mathrm{typecport}))$; $\neg(\mathrm{ProducedBefore}(\text{l-2021}, \mathrm{yr2010}) \oplus \mathrm{ProducedBy}(\text{l-2021}, \mathrm{lg}))$. **Conclusion:** $\mathrm{Have}(\text{l-2021}, \mathrm{typecport}) \oplus \mathrm{ProducedBy}(\text{l-2021}, \mathrm{lg})$.

- Asserting the negated conclusion forces Have $\leftrightarrow$ ProducedBy. Together with ProducedBefore $\leftrightarrow$ ProducedBy, unit propagation gives Have $\leftrightarrow$ ProducedBefore.
- If Have is true this contradicts the first premise; if it is false it contradicts premises 2 and 3. The empty clause $\square$ is derived.
- Prediction: `True`.

5 Limitations

Inference is closed-world, so non-derivable statements are labelled `False`; FOLIO `Uncertain` cannot be distinguished from `False`. The DPLL solver is step-bounded, which trades recall of true entailments for speed.